strategy, results of operations, or financial condition.[2781] For any risks identified, registrants are required to provide information necessary to an understanding of the nature of the risk presented and whether the risk is a physical or transition risk. Registrants are also required to classify whether these risks are reasonably likely to manifest in the short-term and in the long-term. For both physical and transition risks, registrants are required, as applicable, to provide detailed information on these risks (e.g., the particular type of transition risk as well as the geographic location and nature of the properties, processes, or operations subject to the physical risk).

This aspect of the final rules will improve investors' understanding of what a registrant considers to be the relevant short-term and long-term climate-related risks that have materially impacted or are reasonably likely to have a material impact on its business. As a number of commenters have noted, climate-related risks often translate into material financial risks with implications for firm growth and profitability, and therefore investors would benefit from a disclosure regime that requires registrants to provide information on climate-related risks that is accurate and more comparable to each other.[2782]

2781 *See* 17 CFR 229.1502(a).

2782 *See, e.g.*, letters from PIMCO ("[W]e believe climate risks often pose a material financial risk, and therefore, investors need disclosure of climate risks that is complete, reliable, and consistent in order to analyze how climate-related risks may affect a company's business or overall financial performance."); Wellington ("Accurate and comparable information about climate risk is critical to Wellington Management's ability to make informed investment decisions on behalf of our clients. Because climate change will continue to profoundly impact society, economies and markets, investors need more information to better price these risks and fully assess the value of an issuer's securities."); and AllianceBernstein ("[M]aterial risks and opportunities associated with climate change as fundamental financial factors that impact company cash flows and the valuation investors attribute to those cash flows. Regulatory changes, physical risks, and changing consumer decision criteria and preferences are all factors that asset managers need to understand and integrate into their investment processes to make optimal investment decisions on behalf of their clients.")

Academic research has found that retail investors as well as institutional investors value and utilize information on climate-related risks in decision-making.[2783] As numerous commenters stated,[2784] climate-related risks and their impacts on businesses are often not reported in a way that is useful to investors. Commenters noted that with the limitations to the currently available climate-related disclosures, extensive costs in the form of data gathering, research and analysis are needed to process them and to fill data gaps where possible in forming investment decisions.[2785] We expect the final rules to reduce these information processing costs for investors.[2786]

2783 *See supra* section IV.B.1. One commenter suggested that institutional investors and retail investors may have different preferences for climate-related information, especially when the former consider investment portfolios and the latter consider individual companies. *See* letter from Society for Corp. Gov. The commenter further argued that retail investors are unlikely to care about climate-related information given their investment horizon. Because of the documented impact of climate-related risks, including distant ones, on asset prices, we disagree with these assertions.

2784 *See* section IV.B; *see also, e.g.*, letters from BlackRock ("Compared to the existing voluntary framework, the Commission's detailed analytical and disclosure roadmap . . . is more likely to increase the comparability and consistency of issuers' climate-related disclosures."); Calvert ("Currently, climate change disclosures are largely voluntary, unverified, and idiosyncratic."); CFA ("The current voluntary climate-related disclosure regime has resulted in inadequate and inconsistent information which falls short of investor demands and prevents market participants from reasonably assessing the risks of climate change."); and Wellington ("Currently, our evaluation of the positive and negative impacts of climate change on issuers is limited by inadequate information and the absence of a standardized framework for disclosure." and that "For a significant number of issuers, information is not sufficient to support equivalent analysis.").

2785 *See* letter from Wellington ("We were able to make these and other determinations based on available information (including internal and external estimates), and only after extensive research and analysis. For a significant number of issuers, information is not sufficient to support equivalent analysis."); Boston Trust Walden ("Evaluation of climate risk across investment portfolios represents a cost to investors and results in the gathering of data that is often incomplete and not comparable. At Boston Trust Walden, our analysts examine quantitative and qualitative climate-related corporate disclosure to enhance our understanding of the existing and potential financial outcomes associated, ranging from risks (e.g., losing the license to operate) to opportunities (e.g., generating new sources of revenue). In the absence of mandated disclosure requirements, we rely on the data of third-party research providers, which includes a mix of issuer provided data and estimates. Our analysts then seek to fill data gaps through additional research and analysis, outreach via written requests, meetings, and shareholder resolutions seeking the expanded disclosure we require. These processes for gathering necessary climate-related disclosures are inefficient and resource intensive.").

2786 The final rules may also lead to a lower cost of capital for some registrants, as we discuss below.

We expect the final rules will help investors gain a more accurate and complete understanding of the climate-related risks that a registrant determines have materially impacted or are reasonably likely to materially impact its strategy, results of operations, or financial condition. By distinguishing between climate-related risks that manifest in the short-term and long-term, the final rules will help inform investors about which risks are salient to their investment decision-making and which are not, depending on the time horizon investors are focused on. For instance, longer term risks may be less certain and are less likely to have impacts on cash flows in the short-term. As such, some investors may choose to focus more on short-term risks. Conversely, an investor with a long investment horizon may choose to focus on the risks that match its investment horizon.[2787] This temporal standard is consistent with an existing MD&A disclosure requirement and therefore should provide a degree of familiarity to registrants and investors as they prepare and analyze these disclosures.[2788] This aspect of the final rules will impose additional costs on registrants (e.g., direct compliance costs and indirect costs resulting from, for example, increased litigation risk). These costs are discussed in greater detail in sections IV.C.1 and IV.C.3.

b. Disclosure Regarding Impacts of Climate-Related Risks on Strategy, Business Model, and Outlook

The final rules require registrants to describe the actual and potential material impacts of any climate-related risks identified in response to Item 1502(a) on the registrant's strategy,

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2787 *See* letter from Vanguard ("climate risks to be material and fundamental risks for investors and the management of those risks is important for price discovery and long-term shareholder returns.").

2788 *See, e.g.,* letter from ABA ("We believe that climate-related matters should be addressed within the same time short- and long-term time frames used in MD&A."); 17 CFR 229.303(b)(1) ("Analyze the registrant's ability to generate and obtain adequate amounts of cash to meet its requirements and its plans for cash in the short-term (i.e., the next 12 months from the most recent fiscal period end required to be presented) and separately in the long-term (i.e., beyond the next 12 months)."); *see also* section II.C.2.

business model, and outlook.[2789] With respect to their strategy, business model, and outlook, the final rules specify that registrants are required to assess, as applicable, any material impacts on a non-exclusive list of items: business operations; products or services; suppliers, purchasers, or counterparties to material contracts (to the extent known or reasonably available); activities to mitigate or adapt to climate-related risks; and expenditure for research and development. Registrants are also required to discuss whether and how the registrant considers these impacts as part of its strategy, financial planning, and capital allocation.

We expect the resulting disclosures to provide investors with a better understanding of how climate-related risks have materially impacted or are reasonably likely to have a material impact on the registrant. Such disclosures will directly benefit investors who use this information to evaluate the financial prospects of the firms in which they are looking to invest.[2790] Discussions of material impacts on strategy, business model, or outlook will help investors determine whether and how registrants are addressing identified material climate-related risks. This type of disclosure could be particularly useful when comparing the approaches taken by similarly situated registrants. For example, one registrant may disclose that it is actively shifting assets away from exposure to flood zones, while another might disclose that it is investing in such assets as they are considered currently undervalued. These disclosures will allow an investor to choose to invest in the company with climate-related risk strategies that best align with the investor's investment objectives.

Under the final rules, if a registrant has adopted a transition plan to manage a material transition risk, it must describe the plan. The registrant must also provide an annual update about

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2789 *See* 17 CFR 1502(b)-(g).

2790 *See* P. Krueger, Z. Sautner & L.T. Starks, *The Importance of Climate Risks for Institutional Investors*, 33 Rev. of Fin. Stud. 1067 (March 2020); Ilhan *et al*. (2023).

any actions taken during the year under the plan, including how these expenditures have impacted the registrant's financial condition, or results of operations, along with quantitative and qualitative disclosure of material expenditures incurred and material impacts on financial estimates and assumptions as a direct result of the transition plan. We expect these disclosures to provide investors with more complete and reliable information about how registrants plan to address material transition risks. A number of commenters indicated that these disclosures would help investors assess the registrant's approach to managing climate-related risks and achieving its climate-related targets and goals.[2791] This benefit could be reduced if these disclosures provide opportunities for greenwashing. However, we expect this risk to be reduced given that these disclosures will include quantitative and qualitative information on expenditures that are filed with the Commission and are subject to the applicable liability provisions under the Securities Act and Exchange Act. The requirement to provide annual updates should further mitigate these concerns. The updating requirement will be particularly beneficial to investors as it will allow them to analyze the impacts of transition plans on a registrant's operations and financial condition over time.

The requirement to describe quantitatively and qualitatively the material expenditures incurred and material impacts on financial estimates and assumptions as a direct result of the transition plan will help investors better understand a registrant's approach to managing climate-

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[2791] *See, e.g.*, letters from CalPERS; Morningstar; Change Finance; *see also* letter from ICI ("We support [transition plan] disclosure as it would inform investors of the nature of the risks and the company's actions or plans to mitigate or adapt to them."); and the CFA Institute ("We support the Proposed Rule's requirement that a registrant disclose, if it has adopted a transition plan (i.e., a strategy and implementation plan to reduce climate-related risks) as part of its climate-related risk management strategy. We agree with the view that it will facilitate investor understanding of whether the company has a plan and whether it may be effective in the short, medium, and long term in achieving such a transition. Presently, many companies have made net-zero commitments by 2050 but have made little if any disclosures regarding how they plan to get there. This requirement would necessitate that they do so.")

related risks so they have information necessary to assess how those actions have impacted the registrant. Including a quantitative description of material expenditures incurred will discourage boilerplate disclosures and, to some extent, facilitate comparisons across registrants. However, we acknowledge commenters who raised concerns about the difficulties of attributing expenditures to these types of activities.[2792] We recognize that similarly situated registrants may take different approaches in their determination of which expenditures to include and whether to quantitatively or qualitatively identify portions of expenditures specifically tied to these activities. To the extent that registrants take different approaches to identifying such expenditures, the comparability benefits of the disclosure will be diminished. Nevertheless, the qualitative discussion accompanying the disclosures should provide the context necessary for investors to understand the registrant's approach to these activities and provide an assessment of the impact of these activities on the registrant's financial condition.

If a registrant uses scenario analysis to assess the impact of climate-related risks on its business, results of operations, or financial condition, and if, based on the results of such scenario analysis, the registrant determines that a climate-related risk is reasonably likely to have a material impact on its business, results of operations, or financial condition, the registrant must describe each such scenario. This description must include a brief description of the parameters, assumptions, and analytical choices used, as well as the expected material impacts on the registrant under each such scenario. Disclosures about the use of scenario analysis to stress test businesses across a range of possible future climate and climate policy scenarios can vary

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[2792] *See supra* notes 1891 and 1892, and accompanying text.

significantly.[2793] As such, the final rules will inform investors about whether a registrant is using scenario analysis to manage a material climate risk, and for those investors who view scenario analysis as an important tool for climate risk management, allow them to factor this information into their investment decisions.[2794] The required disclosures around parameters, assumptions, and analytical choices used by a registrant when conducting scenario analysis will allow investors to better understand the methodology underlying the scenario analysis and thereby improve investors' assessment of the appropriateness of a registrant's strategy and business model in light of foreseeable climate-related risks.[2795]

If a registrant's use of an internal carbon price is material to how it evaluates and manages climate-related risks disclosed in response to Item 1502(a), then the registrant must disclose in units of the registrant's reporting currency information about the price per metric ton of CO_2e, and the total price, including how the total price is estimated to change over the short-term and long-term, as applicable. For registrants that use more than one internal carbon price to evaluate and manage a material climate-related risk, these disclosures apply to each internal carbon price and the registrant must disclose reasons for using different prices. If the scope of entities and operations involved in the use of an internal carbon price described is materially different from the organizational boundaries used for the purpose of calculating GHG emissions

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2793 *See Using Scenarios to Assess and Report Climate-Related C2ES 20 Financial Risk,* C2ES (Aug. 2018), https://www.c2es.org/document/using-scenarios-to-assess-and-report-climate-related-financial-risk/; FRB of New York, *Climate Stress Testing*, Staff Report No. 1059 (2023).

2794 *See*, *e.g*., Council of Institutional Investors; Boston Common Asset Management; Boston Walden Trust; Domini; University Network for Investor Engagement; AllianceBernstein.

2795 *See, e.g*., letter from Bloomberg (stating "scenario analysis is a useful tool to describe the resilience of a company's strategy to the risks and opportunities of climate change and to develop a more informed view of implications for enterprise value and value chains"); *see also supra* notes 540-542 and accompanying text; *see also* letter from Wellington ("[i]nformation concerning scenario analysis would also help investors evaluate the resilience of the registrant's business strategy in the face of various climate scenarios that could impose potentially different climate-related risks.").

pursuant to Item 1505, the final rules require registrants to describe the difference. We expect this disclosure will provide investors with more standardized and decision-useful information regarding whether a registrant's use of an internal carbon price is material and, if so, how it impacts its strategy, results of operations, and financial condition. This is important to address issues with increased voluntary corporate disclosures of internal carbon pricing.[2796] By mandating that registrants disclose any material differences in their boundaries used for internal carbon pricing and GHG emissions measurement, the final rules will help clarify for investors the scope of entities and operations included in a registrant's application of internal carbon pricing and improve the transparency about the methodology underlying the use of internal carbon pricing so that investors may better compare such use across registrants.

In addition to the general cost considerations discussed in section IV.C.1.b, these provisions may have certain unintended effects on registrants and investors. In particular, as some commenters noted, it is possible that requiring registrants to disclose specific facts about their use of transition plans, scenario analysis, and internal carbon prices to address climate-related risks could deter registrants from utilizing these methods or cause them to abandon them, for example because of perceived litigation risk or because of the direct costs of preparing such disclosure.[2797] This could have negative consequences for investors if the use of these methods would have helped registrants better manage climate-related risks and therefore make value-maximizing decisions in light of those risks. However, if registrants' use of these methods

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[2796] S*ee* CDP, *supra* note 608.

[2797] *See, e.g.*, letters from OMERS; Cemex; and NAM.

becomes a common practice,[2798] due to investor demand or otherwise, this deterrence effect is likely to be limited.

There are potential costs that could result from scenario analysis disclosures under the final rules. First, commenters expressed concern that the disclosure of the scenario analysis results could confuse investors to the extent they inadvertently suggest that the chance of a loss occurring due to a rare event is more likely.[2799] The commenters' concern could materialize if, for instance, a scenario analysis suggests a heightened risk of a once-in-a-hundred-year flood over the next 30 years, and disclosure of this causes certain investors, particularly those not familiar with such analysis, not to invest in the registrant despite the fact that the registrant actually has the same risk profile as other companies that have not made this disclosure. However, we expect any potential investor confusion in such a case will be mitigated because, under the final rules, the registrant would not be required to disclose this information if it determines that this scenario, like other very remote scenarios, are not likely to have a material impact on its business or financial condition. In addition, when disclosure is required, information accompanying the scenario analysis results—such as the assumptions and parameters underlying the analysis—should help provide investors the necessary context for understanding the import of the disclosed analysis.[2800] Second, in disclosing scenario analysis assumptions and inputs as well as information about internal carbon prices, a registrant may face competitive harm to the extent that the disclosures reveal competitively sensitive information,

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2798 S*ee, e.g.,* Climate Action 100+, *Progress Update 2022* (2022), available at https://www.climateaction100.org/wp-content/uploads/2023/01/CA-100-Progress-Update-2022-FINAL-2.pdf (stating that "91% of focus companies have now aligned with TCFD recommendations, either by supporting the TCFD principles or by employing climate scenario planning").

2799 *See, e.g.*, letter from the BPI.

2800 We note that other disclosure requirements, such as those relating to market risk disclosures, convey to investors complex information about uncertain future risks that registrants face.

such as asset allocation decisions. However, we expect that the degree of flexibility offered by the disclosure requirements in the final rules will help avoid the exposure of confidential or proprietary information, though they may make the disclosures less comparable.

Overall, by focusing on climate-related risks that are material to the registrant's business, the final rules seek to avoid imposing costs associated with disclosing large amounts of detailed information that may be less relevant to investors. Finally, some of the required disclosures (*e.g.*, forward-looking statements concerning transition plans, scenario analysis, and internal carbon pricing) will be subject to PSLRA safe harbors, which may reduce litigation costs where the safe harbors are applicable.[2801]

c. Governance Disclosure

The final rules require a registrant to disclose information concerning the board's oversight of climate-related risks as well as management's role in assessing and managing the registrant's material climate-related risks.[2802] The final rules require a registrant to identify, if applicable, any board committee or subcommittee responsible for the oversight of climate-related risks and to describe the processes by which the board or such committee or subcommittee is informed about such risks. Additionally, if there is a disclosed climate-related target or goal or transition plan, the registrant must describe whether and how the board oversees progress against the target or goal or transition plan. In describing management's role in assessing and managing the registrant's material climate-related risks, the registrant should address, as applicable, the following non-exclusive list of disclosure items: (1) whether and which management positions or committees are responsible for assessing and managing climate-related risks and the relevant

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2801 *See supra* section II.J.3 for a discussion of the disclosures required under the final rules that will be subject to PSLRA safe harbors. *See also* 17 CFR 229.1507.

2802 *See* 17 CFR 229.1501.